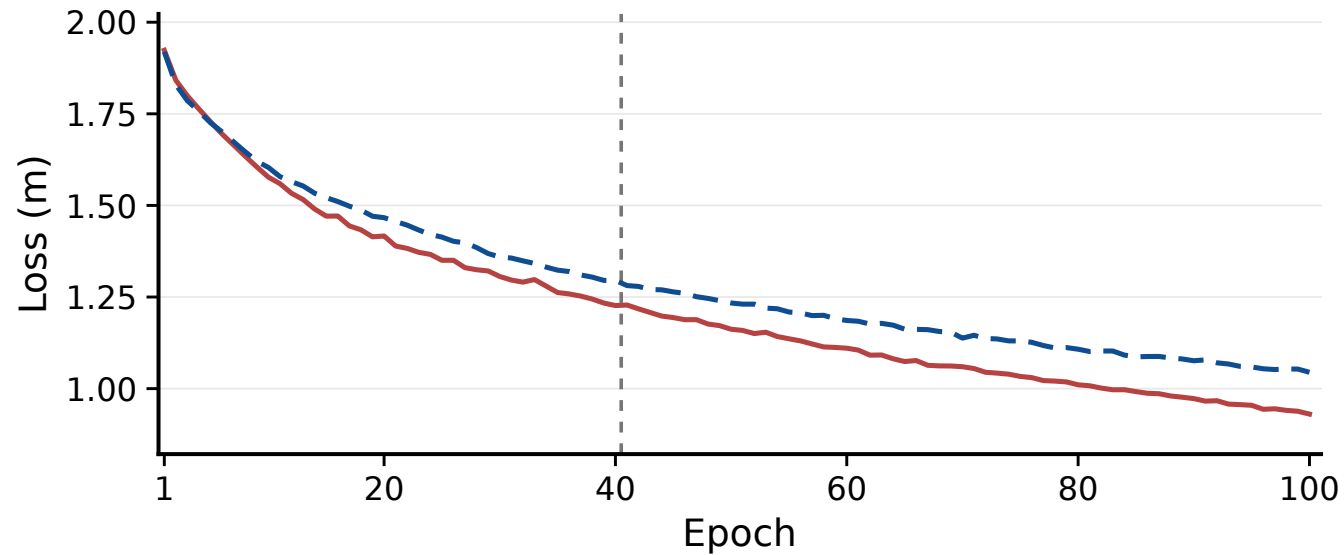


Training history · 100 epochs · seed 2026

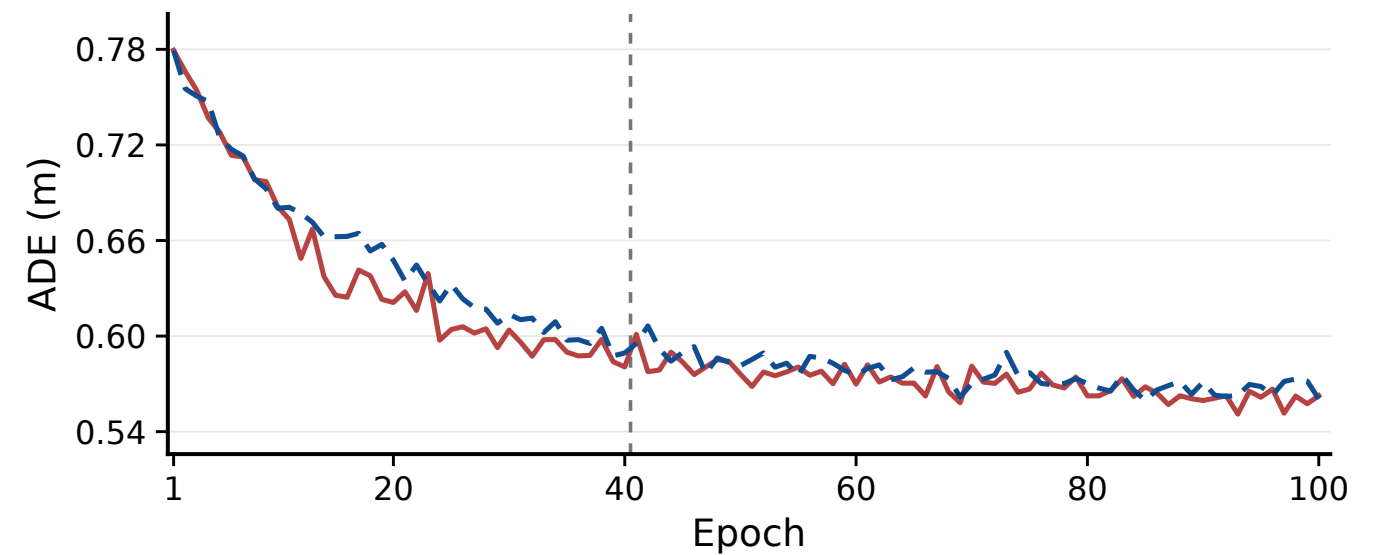
Same global batch 16; execution settings switch between epochs 40 and 41

— GNN - - - Relation QGAT - - - - 40 → 41 execution switch

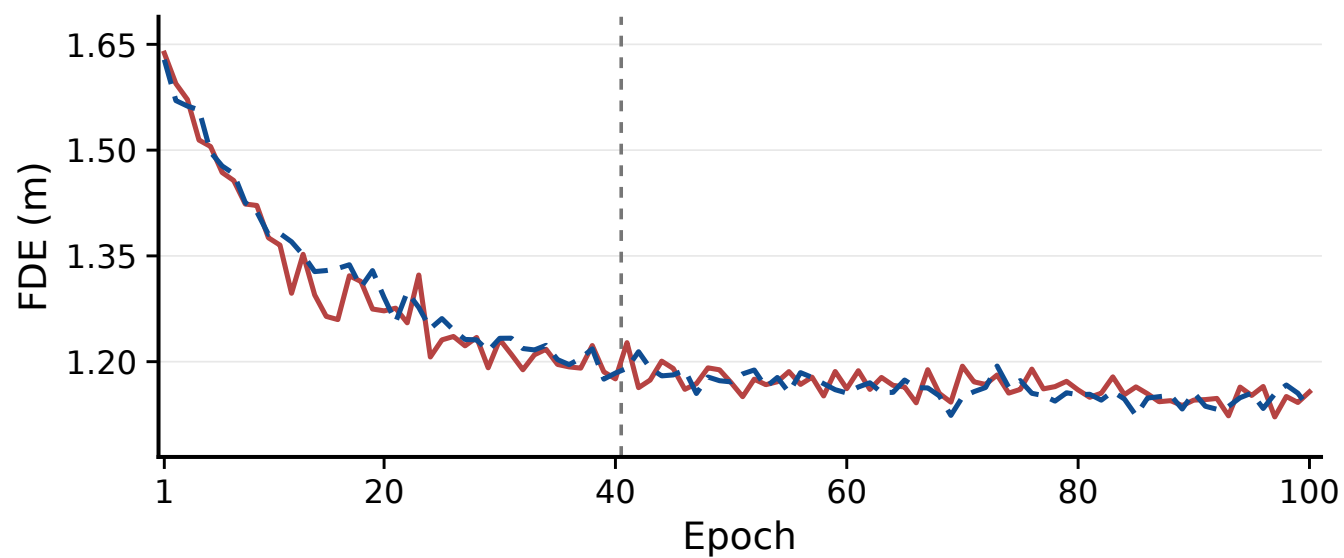
(a) Training loss



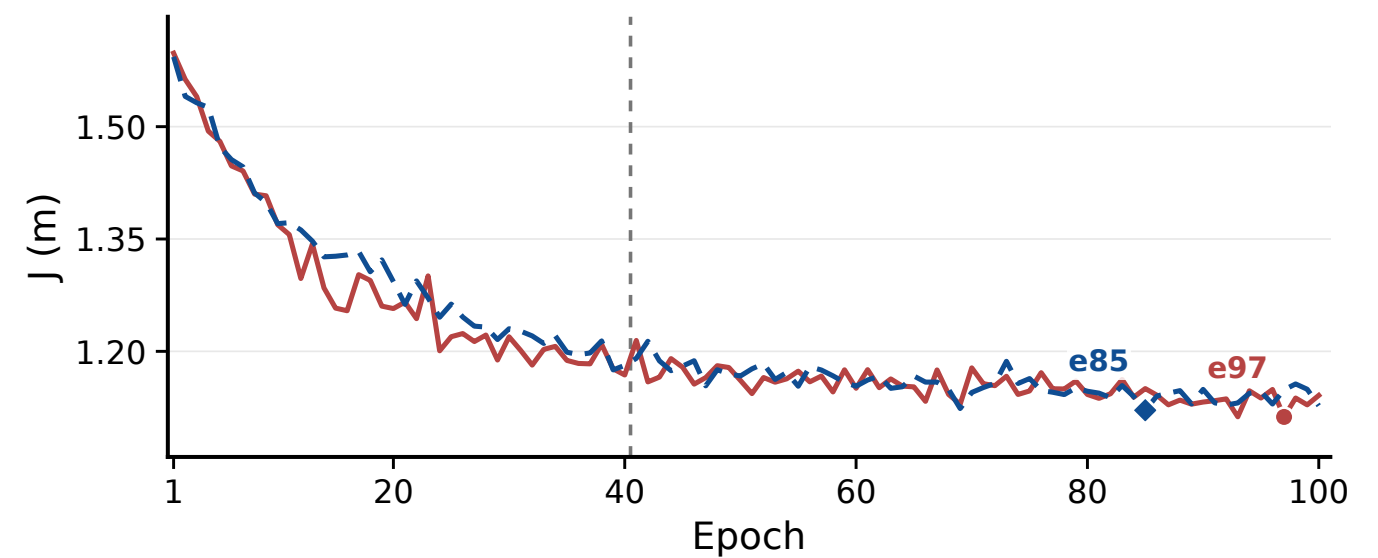
(b) Validation ADE



(c) Validation FDE



(d) Validation J



Best validation J: GNN e97 = 1.112432 m | Relation QGAT e85 = 1.121324 m

Raw epoch values; no smoothing. Validation uses the same 400 origins at four SNRs.